

Qays Mohamed Abubakr Layas

Date of birth: 4 / 4 / 1976

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Qualification: Master degree of electrical engineering from university of Colorado at Denver, USA.

Major: power

Expertise: electrical machines protection and control systems.

Academic experience: designing and planning of substations and renewable energy systems (wind and solar energy systems).

SUMMARY

- Institution of industries and energy of Libya , 5 years of experience (from 2002 to 2007).
- Practical experience about power distribution system and power transformers for Derna city and neighbors.
- Libya **GECOL (General Electrical Company Of Libya)**, **20** years of experience (from 2004 to present).
- 5 years lecturer assistant in **FACULTY OF QAMINIS** (since 2019 to present)
- Practical experience related to the eastern network for faults analysis and system protection setting of levels 400,220,66,30 kv
- Strong background related to the designing of the renewable energy systems including researches and IEEE papers
- English language proficiency is high and certified by :
 - 1- English language programs and centers (**ELS**) **Portland, Oregon State, USA.**
 - 2- Intensive English language program (**IELP**) **Portland, Oregon State, USA.**
 - 3- Spring international language center as a second language program (**ESL**) **Denver, Colorado State, USA.**
- Competent with **NEPLAN** applications (for practical purposes).
- Competent with **MathCAD** applications (for academic purposes).

EDUCATION AND TRAINING

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| ➤ Bachelor's degree, B.S., Garyounis University (University Of Benghazi) , Benghazi | achieved in 7/2001 |
| ➤ Siemens technology, training course for SINCAL package for power system analysis, Tripoli | achieved in 10/2005 |
| ➤ Siemens Technology, training course for protective relays programming and setting , Germany | achieved in 7/2006 |
| ➤ Master of Science degree, MSEE, University of Colorado at Denver , USA | achieved in 5/2011 |

WORK EXPERIENCE

- **From 2002 to 2007:** Street lighting and distribution transformers commissioning and supervising of big consumers and industries with Institution of Industries and Energy of Libya .
- **From 2004 to present :** High and medium voltage networks protection, Electrical engineer / analyst of the various faults concerning with the eastern network of Libya with **GECOL**.

➤ **From 2011 to present :**

- Head of programming and faults analysis department for high voltage in Libyan eastern network of **GECOL**
- Machinery systems and operations (motors and generators).
- Power system network protection and relaying system coordination.
- Substation design and equipment sizing.
- Short circuit and load calculations for distribution equipment sizing and for high voltage networks and system design.
- Digital relays programming and protective devices setting.
- **Gecol's** projects and commissioning for substations constructions and testing with **ABB** and **SIEMENS**.
- Researches and studies and IEEE papers regarding of the fuel cells and wind power stations (renewable energy).
- Projects related to digital control systems and blind signal processing systems.
- interests and researches about ground grid design and surge arresters problems in the eastern network by using ETAP software.
- Giving lectures about power system protection and relay coordination to engineers and trainers with **GECOL** training center in Benghazi.
- Giving lectures to **LBU (LIBYAN BRITISH UNIVESITY)** courses : (power system control, control system, digital signal processing) .
- Giving lectures to **FACULTY OF PETROLUM ENGINEERING TECHNOLOGY** courses : (wind energy, power system control, electrical machines, electrical measuring instruments) .
- Giving lectures to **MIDDLE EAST UNIVESITY FOR MODERN SCIENCES** courses : (electrical circuits 2 , electrical machines, control system, digital signal processing) .
- giving lectures to **university of Benghazi** courses : (electrical department and mechanical department and petroleum department) giving courses machines,power transformers , generators and motors, electrical circuits , electronic circuits, digital signal processing, electrical circuits 1 , electrical circuits 2)
- giving lectures to **faculty of QAMINS** courses (electrical circuits 1, electrical circuits 2, electronic circuits2,3, electrical machines, digital signal processing)

ADDITIONAL EXPERIENCE

Softwares

- **Microsoft office suit** (Word, Excel, Visio, Power Point)
- **MathCAD** for academic applications
- **NEPLAN** for practical power system networks operation and simulation , **SINCAL** Siemens package.
- **ETAP** for practical applications

Some of practical courses of master degree (transcript) taken at university of Colorado Denver, USA

- Electric Drive Systems (power electronic and machines)
- Advanced Electric Machinery (power machines)
- Renewable Energy Systems (power)
- Energy Conversion (power)
- Power systems Dynamics-Protection (power)

- Power Systems Analysis (power)

Projects achieved with courses of master degree

1. Limiting The Single Phase Fault Current By Grounding Resistance (power protection system , grounding technology).
2. Analysis of Bloom Energy ES-5000 and Bloom Energy Claims (renewable energy , fuel cell technology).
3. Adaptive control system and data estimation

